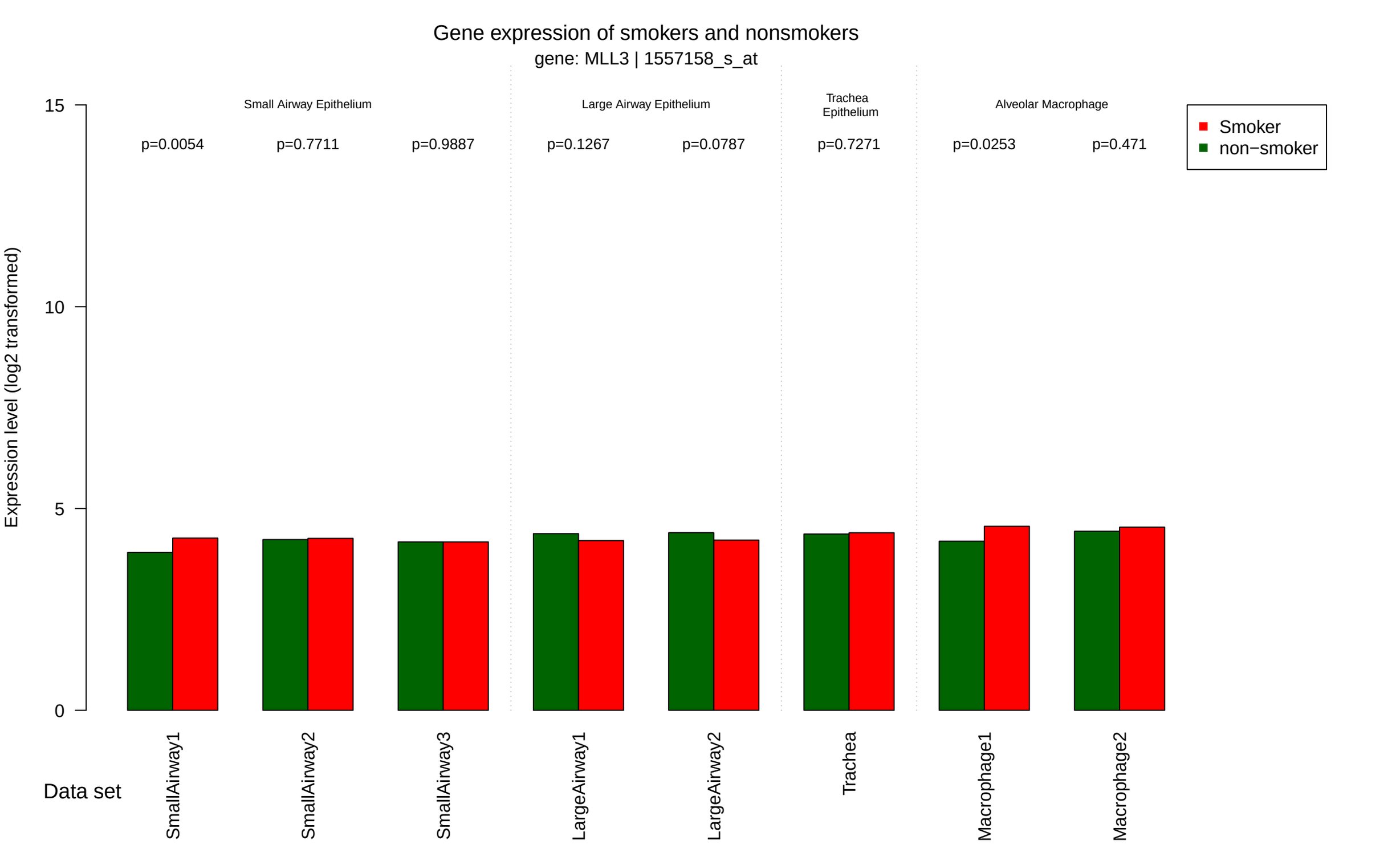
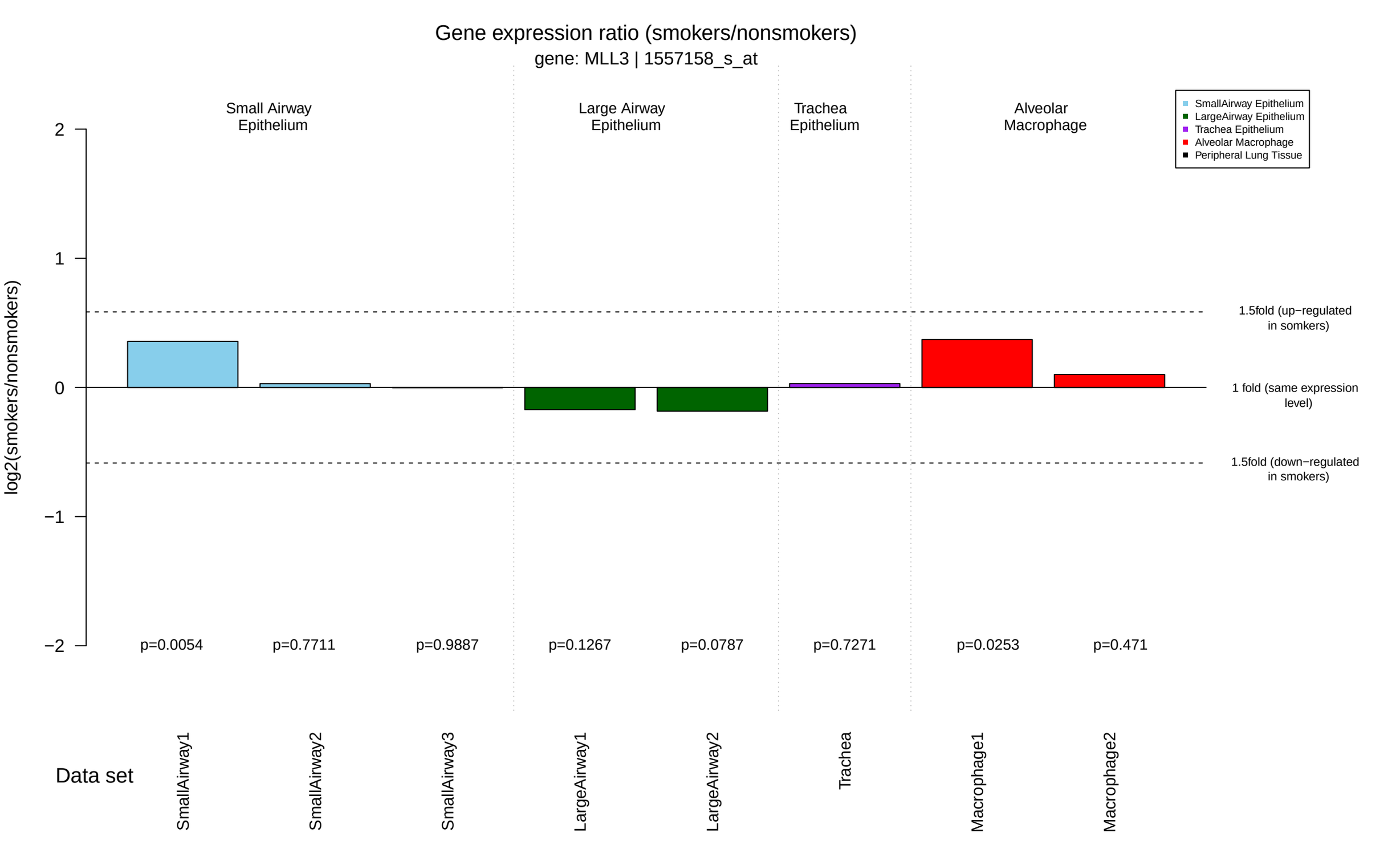
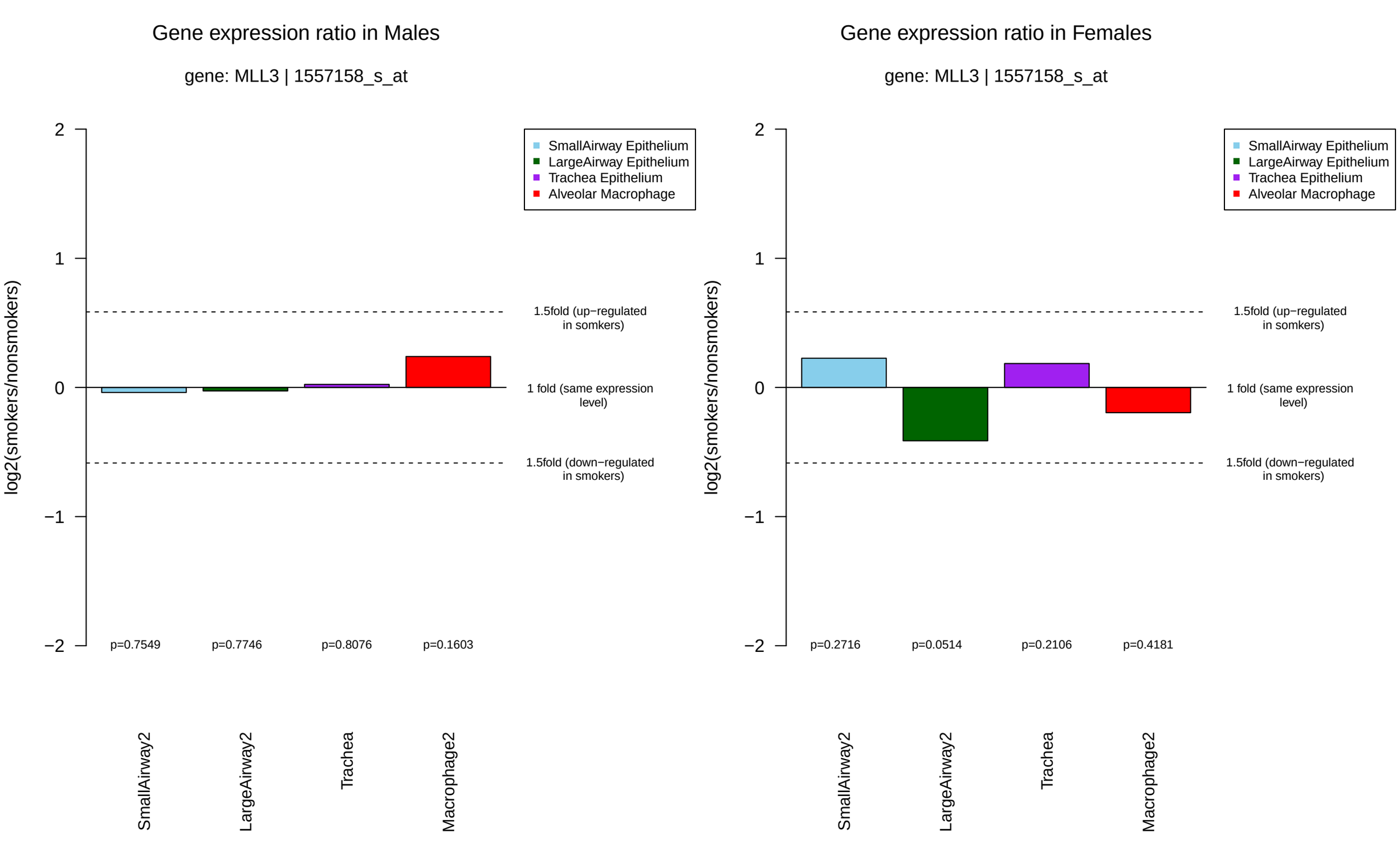
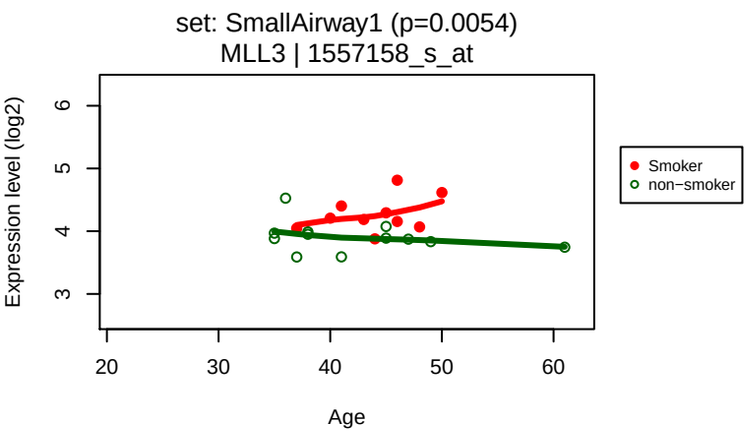
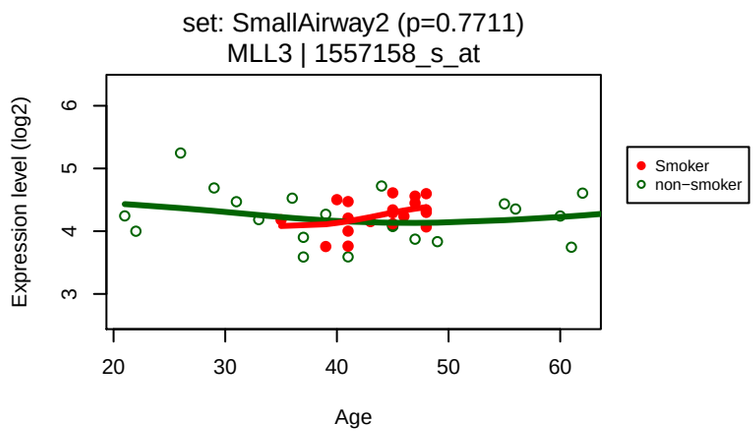
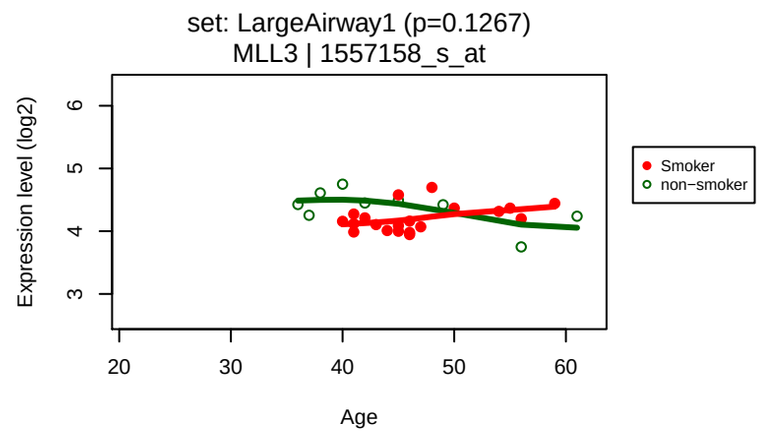

Gene expression of smokers and nonsmokers
gene: MLL3 | 1557158_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0054
p=0.7711
p=0.9887
p=0.1267
p=0.0787
p=0.7271
p=0.0253
p=0.471
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

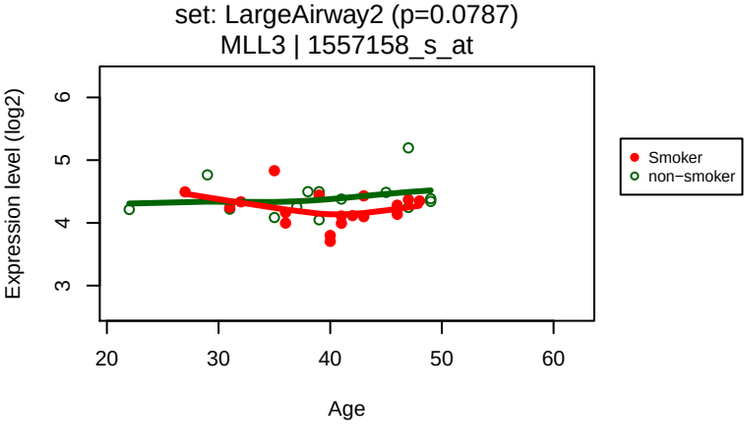

Gene expression ratio (smokers/nonsmokers)
gene: MLL3 | 1557158_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0054
p=0.7711
p=0.9887
p=0.1267
p=0.0787
p=0.7271
p=0.0253
p=0.471
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

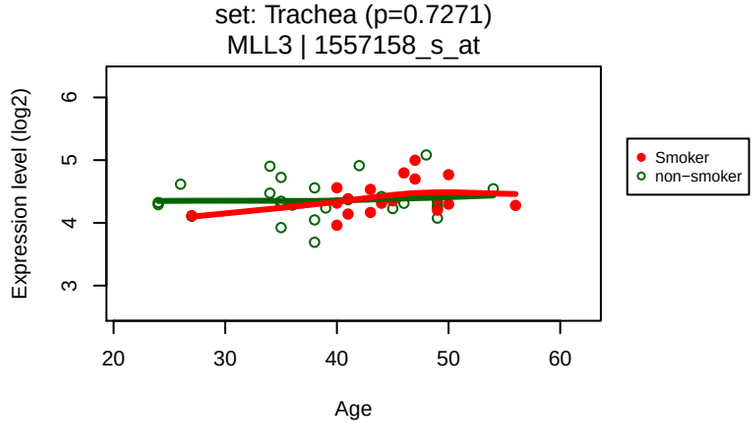

Gene expression ratio in Males
gene: MLL3 | 1557158_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7549
p=0.7746
p=0.8076
p=0.1603
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MLL3 | 1557158_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2716
p=0.0514
p=0.2106
p=0.4181
SmallAirway2
LargeAirway2
Trachea
Macrophage2

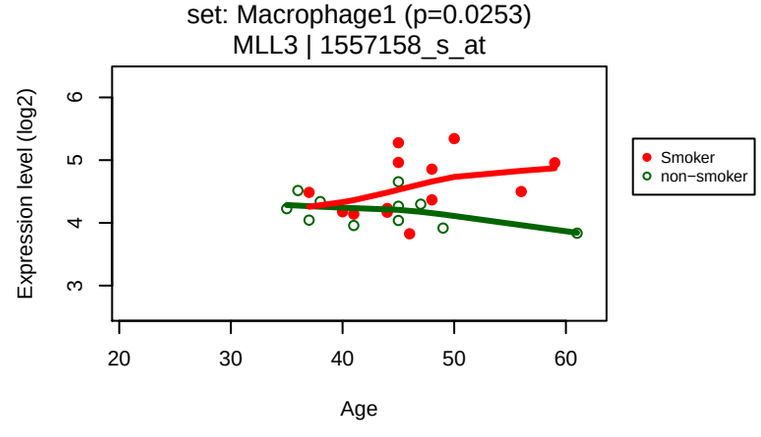

set: SmallAirway1 (p=0.0054)
MLL3 | 1557158_s_at
Expression level (log2)
Age
Smoker
non−smoker

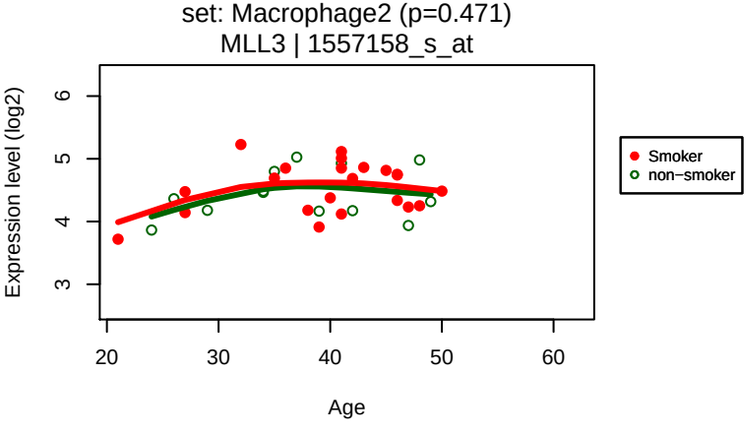

set: SmallAirway2 (p=0.7711)
MLL3 | 1557158_s_at
Expression level (log2)
Age
Smoker
non−smoker

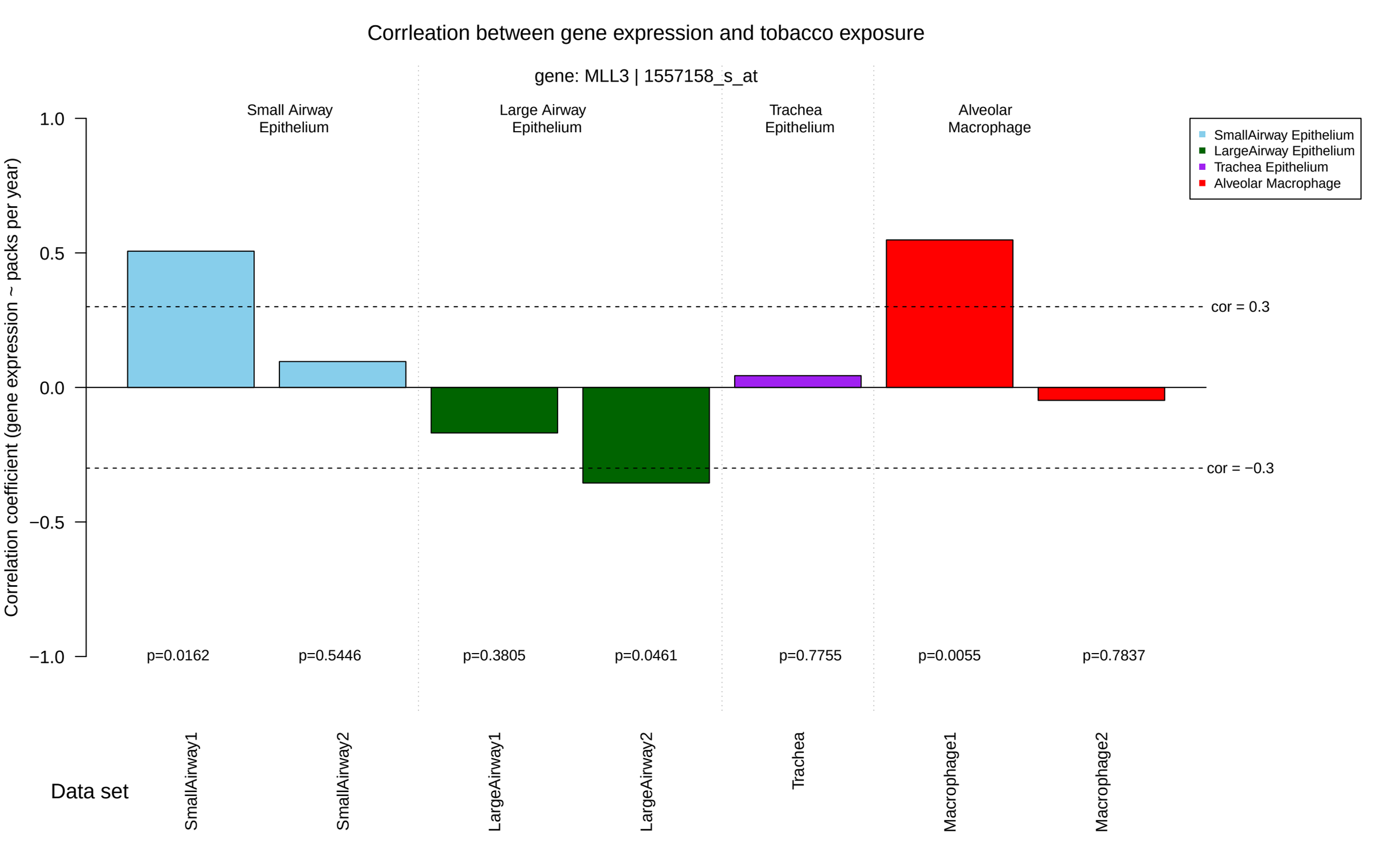

set: LargeAirway1 (p=0.1267)
MLL3 | 1557158_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0787)
MLL3 | 1557158_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7271)
MLL3 | 1557158_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0253)
MLL3 | 1557158_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.471)
MLL3 | 1557158_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MLL3 | 1557158_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.0162
p=0.5446
p=0.3805
p=0.0461
p=0.7755
p=0.0055
p=0.7837
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2